THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A compass needle placed below a wire carrying current from south to north is deflected. If the current is reversed, the deflection 1

(a) becomes zero **(b)** doubles **(c)** stays the same **(d)** is reversed

2. A straight conductor carrying current from east to west is placed in a magnetic field pointing down. The force is directed 1

(a) to the south **(b)** to the north **(c)** downward **(d)** upward

3. If a magnet is moved faster into a coil, the induced current 1

(a) remains the same **(b)** decreases **(c)** becomes zero **(d)** increases

4. The magnetic field produced by a straight wire at a distance 4 cm is B. At 2 cm from the wire (same current), the field is 1

(a) zero **(b)** equal to B **(c)** smaller than B **(d)** larger than B

5. An alpha particle moving north is deflected towards the east in a magnetic field. The force is 1

(a) along the velocity **(b)** against the velocity
(c) zero **(d)** perpendicular to the velocity

6. The induced current in a coil is zero when 1

(a) the magnet is moved
(b) the coil is moved
(c) the field changes
(d) the magnet is at rest relative to the coil

7. Looking at a circular coil, the current is anticlockwise. The face seen behaves as 1

(a) a south pole **(b)** neutral **(c)** a north pole **(d)** both poles

8. A conductor of length L carrying current I is placed parallel to the field. The force is 1

(a) half of maximum **(b)** equal to the weight
(c) maximum **(d)** zero

9. A coil with 100 turns instead of 50 turns, other things same, gives 1

(a) the same induced emf **(b)** no emf
(c) a smaller induced emf **(d)** a larger induced emf

10. When current in a solenoid is doubled (other factors same), the field inside 1

(a) increases **(b)** becomes zero **(c)** decreases **(d)** remains the same

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State why two field lines cannot intersect each other. 2

12. A current-carrying wire is parallel to a uniform magnetic field. State the force and give the reason. 2

13. How many times does the current of frequency 50 Hz change direction in 1 minute? 2

14. A coil carries current clockwise as seen from the left. Which pole is at the left face? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bar magnet is moved towards a coil connected to a galvanometer and then away. Describe the observations and give the reason. 3

16. State differences between a DC generator and an AC generator. 3

17. What is the difference between the magnetic field due to a straight wire and that due to a solenoid? 3

OR

How does the magnetic field due to a solenoid vary 3
(a) with the current
(b) with the number of turns per unit length
(c) with a soft iron core?

18. Why does the induced current in a generator last only during rotation? 3

OR

Why are the house wiring and appliances connected in parallel but a fuse in series? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the magnetic field pattern around a current-carrying straight wire and how it changes when the current is doubled and when the distance from the wire is doubled. Describe how to show the direction of the field using the right-hand thumb rule. 5

OR

A household circuit has a 5 A circuit for lights and fans and a 15 A circuit for power appliances. Describe its layout, state why a geyser of 2 kW needs the 15 A circuit and why a fuse is used. 5

20. A student wants to make a strong electromagnet. Describe how it is made, state the factors on which its strength depends and state why it is called a temporary magnet. 5

OR

Explain what is meant by short circuit and overloading in household wiring. Give two causes of each and mention three precautions. 5

END OF QUESTION PAPER — Total: 40 Marks